

Hey everyone,

I was looking at ways I could contribute to the project without overlapping too much with the current development work. I found a few backend and support modules that could add real value to the MVP and make the system more robust.

Potential Contributions:

1. Conflict Detection Engine

- Prevent overlapping bookings for the same facility.
- Ensure booking requests are validated before approval.

2. Booking State Management

- Handle valid booking transitions such as:
 - Pending → Approved
 - Pending → Rejected
 - Approved → Cancelled
- Prevent invalid status changes and maintain consistency.

3. Slot Transfer / Adjustment Service

- Support the Adjustment & Relinquish feature.
- Ensure booking ownership transfer happens safely and atomically when a user accepts an adjustment request.

4. Analytics Module

- Generate reports such as:
 - Total booked hours
 - Cancellation rate
 - Peak demand periods
 - Most utilized facilities
- Can be used directly in the admin dashboard.

5. Audit Logging

- Track all major actions (booking, approval, rejection, cancellation, adjustment requests).
- Useful for accreditation, maintenance records, and debugging.

6. Dummy Data / Seed Scripts

- Generate facilities, bookings, users, and adjustment requests for testing and demos.
- Helps reviewers quickly test the system.

7. Role-Based Access Control

- Define permissions for Students, Faculty, and Admins.
- Ensure only authorized users can approve requests or access analytics.

I think these modules would strengthen the project technically while also helping us during the final demonstration and defense. I'm happy to take ownership of any of these areas based on what the team feels is most needed.